# Robustness Evaluation of Large Language Models on GSM8K via Controlled Mathematical Variations

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**Abstract**

Large language models (LLMs) have demonstrated strong performance on mathematical reasoning benchmarks such as GSM8K. However, their robustness to small perturbations in problem formulation remains underexplored. In this work, I propose a systematic evaluation framework that measures the robustness of reasoning models under controlled variations of math word problems. Starting from 100 base problems sampled from the GSM8K dataset, I generate five types of semantic and structural variations, including semantic rephrasing, irrelevant numerical distractors, symbolic or operand swaps, scale or unit changes, and implicit reasoning requirements. I evaluate several modern LLMs using both Chain-of-Thought (CoT) and Self-Consistency (SC) inference strategies. My experiments reveal that while strong models maintain high baseline accuracy, performance degradation occurs under certain perturbations, particularly symbolic transformations and scale modifications. Self-consistency decoding improves robustness across most models. These findings highlight important limitations in mathematical reasoning robustness and provide insights into how reasoning models respond to adversarial variations in problem statements.

## 1 Introduction

Large language models (LLMs) have recently achieved strong performance on mathematical reasoning benchmarks, especially on datasets such as GSM8K [1]. In addition, prompting strategies such as Chain-of-Thought (CoT) [2] and inference-time aggregation strategies such as Self-Consistency (SC) [3] have substantially improved answer accuracy on multi-step reasoning tasks.

Despite these advances, benchmark accuracy alone does not guarantee that a model is reasoning in a robust and generalizable way. A system may answer the original version of a question correctly yet fail after a mild rephrasing, an irrelevant distractor, or a unit transformation. Such failures are especially

important in mathematical reasoning, where small surface changes should not affect the underlying solution if the problem semantics remain equivalent.

This project studies robustness on GSM8K-style problems through a controlled variation framework. Starting from 100 base GSM8K problems, I create five families of variations that preserve the gold answer while modifying the problem form in targeted ways: semantic rephrasing, irrelevant numerical distractors, symbolic or operand swaps, scale or unit changes, and implicit reasoning requirements. I then evaluate several open-weight LLMs under both CoT and SC to analyze how robustness varies by model size, reasoning capability, and inference strategy.

My contributions are threefold. First, I build a controlled robustness benchmark of 600 GSM8K-style items. Second, I provide a careful human verification stage to ensure that transformations preserve the original answer. Third, I report a detailed empirical study across multiple model families, analyzing both absolute accuracy and Perturbation Degradation Rate (PDR), along with variation-specific failure patterns and error distributions.

## 2 Related Work

Mathematical reasoning has become one of the most prominent evaluation settings for modern LLMs. GSM8K [1] established a widely used benchmark for grade-school mathematical word problems requiring multi-step reasoning rather than direct lookup or pattern completion. Since then, a large body of research has focused on improving reasoning performance on such datasets.

A major breakthrough came from Chain-of-Thought prompting [2], which showed that prompting models to produce intermediate reasoning steps can substantially improve performance on arithmetic, symbolic, and commonsense reasoning tasks. Self-Consistency [3] further improved this approach by sampling multiple reasoning trajectories and selecting the most frequent final answer. These methods established the importance of inference-time reasoning behavior rather than final-answer generation alone.

More recently, reasoning-focused models and reasoning-oriented training pipelines have become an active research area. DeepSeek-R1 [4] demonstrates that reinforcement learning can significantly improve reasoning performance, while distilled reasoning models make stronger reasoning behaviors accessible at smaller model sizes. These developments suggest that reasoning capability is not only a function of scale, but also of training objective and decoding strategy.

At the same time, robustness-oriented evaluation has grown increasingly important. General work on dataset difficulty and example instability, such as Dataset Cartography [5], highlights that model performance often depends on subtle variations in training or evaluation examples. In mathematical reasoning specifically, robustness has received renewed attention in 2025 and 2026.

Math-RoB [6] introduces a benchmark for reasoning robustness in LLMs, focusing on hallucination-triggering and robustness gaps under controlled problem changes. Yang et al. [7] specifically evaluate robustness to numerical variations

in mathematical reasoning and show that even strong LLMs often fail to preserve valid reasoning under modified numeric forms. In a related direction, Anantheswaran et al. [8] study noisy and adversarial math word problems and show that irrelevant information can significantly degrade model performance.

Research in 2026 also begins to examine robustness at the reasoning-trace level. von Recum et al. [9] investigate how reasoning models respond to interventions on their own chain-of-thought, while Song et al. [10] provide a broader taxonomy of reasoning failure modes in LLMs. These studies reinforce the view that strong accuracy does not imply robust reasoning.

My work is complementary to this literature. Rather than proposing a new training method, I build a controlled evaluation framework over GSM8K-style questions and analyze how different perturbation families affect multiple open-weight models under both CoT and SC. This allows us to study robustness in a setting that is interpretable, reproducible, and closely aligned with standard mathematical reasoning benchmarks.

# 3 Method for Variation

## 3.1 Base Problem Selection

I begin with 100 base problems sampled from GSM8K. These base problems serve as the foundation for all subsequent transformations. The goal is not to create arbitrary adversarial attacks, but to generate controlled variations that preserve the correct gold answer while perturbing the surface form, structure, or reasoning cues of the original question.

Each base question is associated with exactly five transformed versions, yielding a total of 600 items:

- 100 base questions
- 100 semantic rephrasing variants
- 100 irrelevant numerical distractor variants
- 100 symbolic or operand swap variants
- 100 scale or unit change variants
- 100 implicit reasoning variants

## 3.2 Variation Families

**Semantic Rephrasing.** These variants preserve the mathematical content but alter wording, phrasing, and sentence structure. The purpose is to test whether the model depends excessively on specific lexical or narrative forms.

**Irrelevant Numeric Distractors.** These variants inject numbers that are not needed for solving the problem. They are designed to test whether the model can identify relevant quantities and ignore irrelevant numerical content.

**Symbolic or Operand Swap.** These variants alter the symbolic presentation of the problem while preserving the final answer. In practice, I used a mixture of subject renaming and safe reordering or local re-expression. The goal is to probe whether models rely too heavily on memorized templates, entity position, or superficial symbolic form.

**Scale or Unit Change.** These variants transform units or numeric form while preserving the answer. For example, a quantity may be expressed through a unit conversion or an equivalent rescaling. When exact and natural unit substitutions were possible (e.g., minutes to hours), they were preferred. When no clean conversion was available, equivalent numeric rescaling was used. This variation family tests whether models maintain correct proportional reasoning under transformed presentation.

**Implicit Reasoning Requirement.** These variants make the reasoning path less explicit, requiring the model to infer a missing step or hidden assumption from context. Instead of merely restating the same problem, these questions slightly reduce direct cues and require more active inferential reasoning while preserving solvability.

### 3.3 Automatic Generation and Manual Control

The generation pipeline combines automated transformations with manual review. Some transformations can be handled with rules (e.g., parts of scale/unit conversion), whereas others require more context-sensitive rewriting. Because automated generation can easily break the gold answer or introduce ambiguity, all generated items were manually checked and selectively revised. This human validation is not an auxiliary detail; it is a necessary component of a valid robustness benchmark.

## 4 Human Evaluation

Human evaluation played a central role in constructing the final benchmark. Automated transformation alone was insufficient because several variation types can accidentally alter the true solution or produce awkward, under-specified, or semantically misleading questions.

My human review focused on four major aspects.

**Answer Preservation.** For each generated variant, I verified that the final answer remained identical to the answer of the corresponding base problem. This was especially important for symbolic and scale-related transformations,

where a seemingly minor wording change can accidentally alter the underlying arithmetic relation.

**Scale and Unit Consistency.** Scale or unit changes were carefully checked to ensure that all transformations were mathematically coherent. In some cases, automatically generated variants introduced conversions that were technically legal but unnatural or overly confusing. These were rewritten to maintain both correctness and readability. In addition, when a natural unit substitution was not available, equivalent rescaling expressions were used conservatively.

**Implicit Reasoning Quality.** Implicit reasoning variants were revised to avoid repetitive or mechanically phrased additions. Instead of always adding the same kind of instruction (e.g., "assume whole items only"), I made the change more question-aware. For counting, grouping, rate, and money problems, I used different formulations so that the resulting item remained natural while still requiring a slightly deeper inference step.

**Symbolic Variation Safety.** For symbolic or operand swap variants, I deliberately avoided transformations that would change the gold answer. I used a mixture of safe subject renaming and local symbolic rewrites, but only when they preserved the mathematical structure. In cases where a more aggressive transformation risked changing referents or arithmetic relations, the variant was simplified or rewritten.

This manual curation phase improved the final dataset in two ways. First, it ensured correctness, which is essential for meaningful evaluation. Second, it increased the realism of the benchmark by preventing unnatural transformations from dominating the difficulty profile.

# 5 Experimental Setup

## 5.1 Models

I evaluate seven open-weight models spanning different sizes and reasoning profiles:

- Llama-3.2-3B-Instruct
- OpenThinker-Agent-v1
- Qwen2.5-7B-Instruct
- Qwen3-0.6B
- Qwen3-1.7B
- DeepSeek-R1-Distill-Qwen-7B

- Qwen3-4B

These models were selected to cover both reasoning-oriented and general-purpose instruction-tuned architectures, as well as a range of parameter scales.

## 5.2 Inference Settings

I evaluate each model under two inference strategies.

**Chain-of-Thought (CoT).** The model is prompted to solve the problem carefully, give a brief explanation in one to three sentences, and end with exactly one line in the format:

```
FINAL: <answer>
```

This prompt structure makes the output easy to parse while preserving a concise rationale for interpretation.

**Self-Consistency (SC).** For SC, the model generates multiple responses for the same question, and the final answer is determined through majority voting over normalized predictions. In my experiments, SC is applied to the base set and a subset of variation families used for robustness stress testing.

## 5.3 Evaluation Protocol

Under CoT, each model is evaluated on all 600 questions:

- 100 base
- 500 variations

Under SC, each model is evaluated on:

- 100 base
- 300 variations

where the SC variation subset includes the three families that are particularly informative for robustness:

- irrelevant_numeric
- scale_tweak
- symbolic_swap

### 5.4 Metrics

I report:

- **Base accuracy**: accuracy on the original 100 GSM8K problems
- **Variation accuracy**: accuracy on transformed variants
- **PDR (Perturbation Degradation Rate)**:

$$\text{PDR} = \frac{\text{Acc}_{base} - \text{Acc}_{var}}{\text{Acc}_{base}} \times 100$$

A larger positive PDR indicates greater degradation under perturbation. A negative PDR means that the model performed slightly better on a variation set than on the corresponding base set.

I also analyze error counts by model, method, and variation type.

## 6 Results

### 6.1 Overall Accuracy

Figure 1 compares base and variation accuracy under both CoT and SC. OpenThinker-Agent-v1 is the strongest overall model in this study, achieving 0.95 base accuracy and 0.918 variation accuracy under CoT, and 0.96 base accuracy and 0.953 variation accuracy under SC. Qwen2.5-7B also performs strongly, with 0.92 base accuracy and 0.912 variation accuracy under CoT, increasing to 0.95 and 0.917 under SC.

At the other end of the spectrum, Qwen3-0.6B performs substantially worse, with 0.63 base accuracy and 0.642 variation accuracy under CoT, and 0.73 base accuracy and 0.697 variation accuracy under SC. Interestingly, its CoT variation accuracy slightly exceeds its base accuracy, producing a negative overall PDR.

DeepSeek-R1-Distill-Qwen-7B and Qwen3-4B occupy a middle-to-strong tier. DeepSeek-R1-Distill-Qwen-7B improves from 0.81/0.806 (base/variation) under CoT to 0.87/0.873 under SC. Qwen3-4B shows similar robustness, with 0.85/0.854 under CoT and 0.86/0.870 under SC.

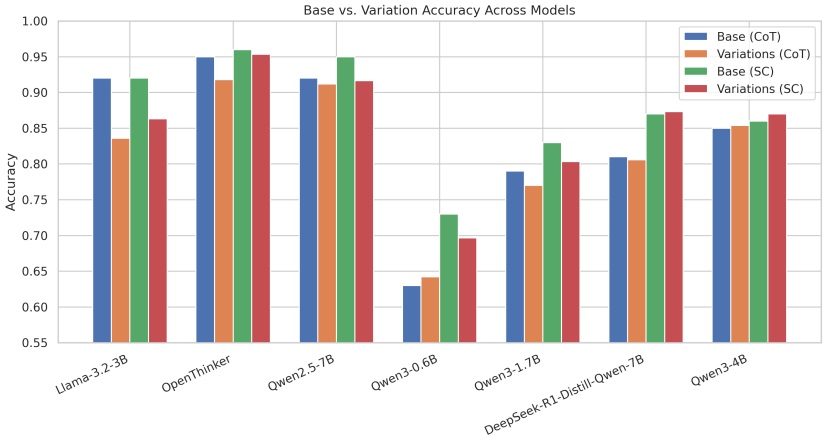


Figure 1: Base and variation accuracy across models under CoT and SC. The y-axis is truncated for readability.

## 6.2 Overall Robustness and PDR

Figure 2 reports overall PDR by model. The most robust model in the full evaluation is OpenThinker-Agent-v1, especially under SC, where its overall PDR drops to 0.694. In contrast, Llama-3.2-3B-Instruct is substantially more sensitive to perturbation, with 9.13% PDR under CoT and 6.16% under SC.

Qwen2.5-7B-Instruct shows strong CoT robustness (0.87% overall PDR), but its SC PDR rises to 3.51%, indicating that while SC improves accuracy, it does not uniformly improve robustness under all perturbations. Qwen3-1.7B also shows modest degradation, with 2.53% PDR under CoT and 3.21% under SC.

Two models exhibit negative overall PDR in some settings: Qwen3-4B under both CoT and SC, and DeepSeek-R1-Distill-Qwen-7B under SC. This indicates slightly higher accuracy on the evaluated variation subset than on the corresponding base set. Such cases likely reflect either sampling effects or the fact that some perturbations do not necessarily increase difficulty for every model.

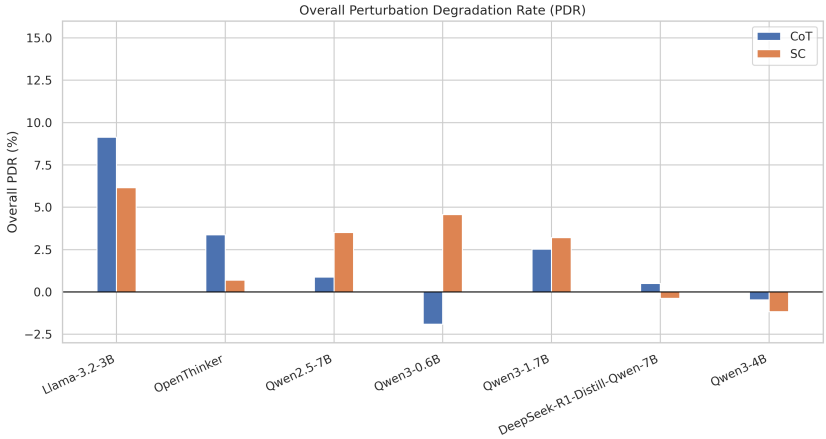


Figure 2: Overall Perturbation Degradation Rate (PDR) across models. Larger values indicate stronger degradation under perturbation. The y-axis is truncated to emphasize relative differences.

### 6.3 Variation-Specific Results

The per-variation results show that not all transformations are equally difficult.

For Llama-3.2-3B-Instruct under CoT, the most difficult variation is `scale_tweak`, with accuracy falling to 0.79 and PDR reaching 14.13%. `symbolic_swap` is also highly disruptive, with 0.82 accuracy and 10.87% PDR. In contrast, under SC, the same model improves on `irrelevant_numeric`, reaching 0.93 accuracy and yielding a negative PDR of −1.09%.

For OpenThinker-Agent-v1, all perturbations are relatively well-handled. Under CoT, its worst family is `symbolic_swap` with 5.26% PDR, while under SC it maintains 0% PDR on both `irrelevant_numeric` and `scale_tweak` and only 2.08% PDR on `symbolic_swap`. This is a strong sign of robust inference behavior.

Qwen2.5-7B-Instruct is also stable under CoT, and in fact shows slightly negative PDR on `critical_thinking` and `symbolic_swap`. However, under SC, `scale_tweak` becomes more difficult, with a PDR of 5.26%.

Among the smaller Qwen3 models, Qwen3-0.6B is the most fragile. Under CoT, `scale_tweak` is its hardest family with 7.94% PDR. Under SC, both `scale_tweak` and `symbolic_swap` remain difficult, with PDR values of 9.59% and 6.85%, respectively. Qwen3-1.7B is stronger, but `symbolic_swap` still emerges as the hardest family, especially under CoT.

DeepSeek-R1-Distill-Qwen-7B performs well overall, but under CoT its `symbolic_swap` family drops to 0.74 accuracy and produces an 8.64% PDR. Under SC, that weakness is largely removed.

Qwen3-4B is an interesting case: it appears highly robust overall, with zero PDR under SC on `scale_tweak` and `symbolic_swap`, and negative PDR on

`irrelevant_numeric`.

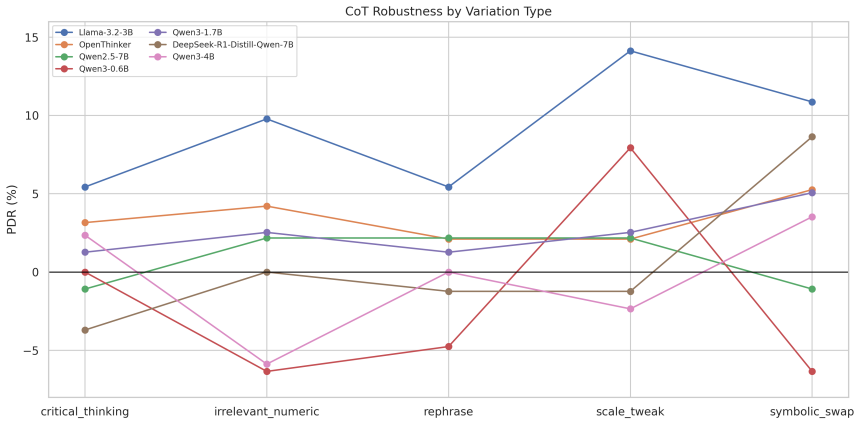


Figure 3: CoT robustness by variation type. Positive PDR means degraded performance relative to the base set. The y-axis is truncated for visual clarity.

### 6.4 Results Summary

Taken together, the results support three broad conclusions:

1. Larger or explicitly reasoning-focused models tend to have higher baseline accuracy and lower robustness degradation.
2. `scale_tweak` and `symbolic_swap` are the most consistently difficult perturbation families.
3. Self-Consistency generally improves performance, but the degree of improvement varies by model and perturbation type.

## 7 Evaluation and Robustness Analysis

This project is not only an accuracy benchmark but also a robustness study. Therefore, interpretation of the results must go beyond simple base accuracy.

### 7.1 Why PDR Matters

Base accuracy answers the question: can the model solve the original problem? PDR answers a different question: if the problem is expressed differently but remains logically equivalent, does the model still succeed? A model with slightly lower absolute accuracy but much lower PDR may be preferable in settings where problem formulations are variable or noisy.

For example, OpenThinker-Agent-v1 is not only the strongest model in raw accuracy, but also one of the most stable under perturbation. In contrast,

Llama-3.2-3B-Instruct has reasonably high base accuracy (0.92), yet its performance degrades substantially under multiple perturbation families, especially scale and symbolic changes. This indicates that high benchmark accuracy can mask fragility.

### 7.2 Why Scale and Symbolic Changes Are Hard

Two variation families stand out repeatedly:

- `scale_tweak`
- `symbolic_swap`

These transformations challenge the model's ability to preserve mathematical structure under altered representation. In scale changes, the model must interpret equivalent quantities across different numeric forms or units. In symbolic changes, the model must maintain correct entity-role alignment despite altered names, local order, or symbolic presentation.

These failures suggest that some models rely partly on shallow heuristics or memorized problem patterns rather than fully abstract relational reasoning.

### 7.3 Effect of Self-Consistency

Self-Consistency typically improves performance because it reduces the influence of any single bad reasoning path. This is especially clear for:

- OpenThinker-Agent-v1
- DeepSeek-R1-Distill-Qwen-7B
- Qwen3-4B

However, SC is not uniformly beneficial for every robustness metric. For example, Qwen2.5-7B-Instruct has stronger absolute base accuracy under SC, but its overall PDR increases relative to CoT. This indicates that increased base performance does not guarantee proportionally improved robustness under all perturbation families.

## 8 Error Analysis

The wrong-count summaries provide an interpretable view of where model failures concentrate.

Figure 4 ranks the most failure-prone model / variation / method combinations. The largest visible wrong count comes from Qwen3-0.6B under CoT on `scale_tweak`, with 42 wrong examples. This is followed by multiple Qwen3-0.6B categories with 33–37 wrong cases, showing that the smallest model is broadly unstable rather than failing only on one specific family.

Llama-3.2-3B-Instruct also exhibits concentrated failures on `scale_tweak` and `symbolic_swap`, consistent with its relatively high PDR. OpenThinker-Agent-v1, by contrast, has much smaller wrong counts, and its SC setting further reduces these counts.

DeepSeek-R1-Distill-Qwen-7B shows one notable weakness under CoT: `symbolic_swap` reaches 26 wrong cases. This is a useful reminder that reasoning-oriented models may still be vulnerable to symbolic presentation changes even when their overall average performance is strong.

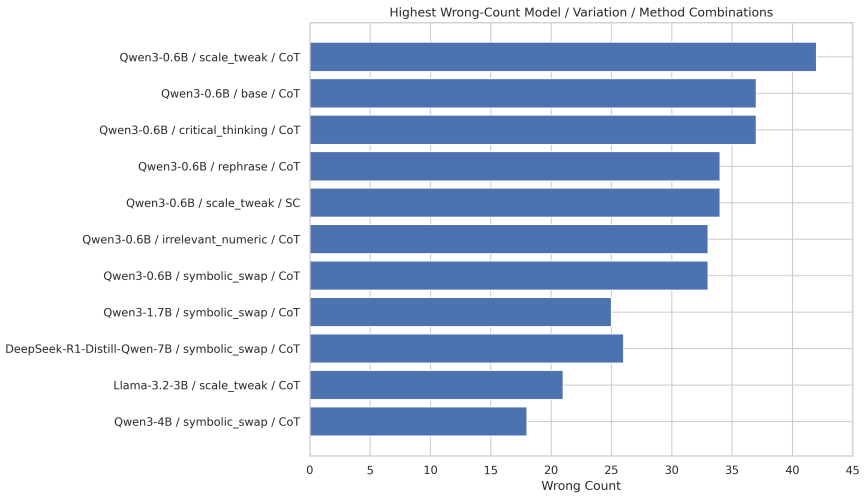


Figure 4: Largest wrong-count model / variation / method combinations. This highlights where failures are most concentrated.

Qualitatively, the failure patterns suggest three recurring modes:

1. **Distractor integration errors**: the model incorporates irrelevant numbers into the solution path.
2. **Scale / unit misalignment**: the model applies the right arithmetic to the wrong numeric representation.
3. **Entity-role confusion**: after symbolic changes, the model assigns a quantity to the wrong person or object.

These are exactly the kinds of failures that a standard benchmark accuracy score cannot diagnose by itself.

# 9 Discussion

The results indicate that mathematical reasoning robustness depends on both model capability and variation type. Larger or more reasoning-specialized mod-

els are not immune to perturbation, but they degrade less sharply and recover better under SC. Meanwhile, smaller models may appear acceptable on the base benchmark while still being highly unstable under simple changes to question form.

An important implication is that future evaluation of reasoning models should include structured perturbations rather than relying solely on original benchmark performance. A model that scores highly on GSM8K but fails under harmless semantic changes may not be reliable in practical settings where users phrase problems in diverse ways.

Another implication concerns inference strategy. SC improves performance for most models, but it should not be treated as a universal fix. In some cases it improves absolute accuracy but leaves substantial robustness gaps, especially on the most challenging perturbation families.

## 10 Limitations

This study has several limitations.

First, the dataset is still relatively small. Although 600 examples are enough to expose meaningful robustness patterns, a larger benchmark would provide more stable estimates across model families and variation types.

Second, the variation generation process includes manual review. This improves correctness but reduces full automation and may introduce some subjectivity in how aggressively variants are rewritten.

Third, SC was evaluated only on a subset of variation families rather than all five. This was a practical choice driven by runtime constraints, but it means direct CoT vs. SC comparison is not perfectly uniform across every family.

Fourth, some plots use truncated y-axes to make differences visually obvious. While this is helpful for presentation, the captions explicitly note that the axes are truncated for readability.

## 11 Conclusion

I introduced a controlled robustness benchmark built from GSM8K by applying five families of mathematically answer-preserving variations. Using this benchmark, I evaluated seven open-weight language models under both Chain-of-Thought and Self-Consistency inference.

The results show that current reasoning models, even when strong on base benchmark accuracy, remain sensitive to structured perturbations. The most difficult transformations are scale/unit changes and symbolic modifications. Self-Consistency generally improves performance, but robustness improvements are uneven across models and variation types.

Overall, this project demonstrates that robustness-aware evaluation reveals important weaknesses that are hidden by standard benchmark accuracy alone.

Future work should expand both the size and diversity of variation-based robustness benchmarks and explore training or decoding methods that explicitly improve invariance to mathematically equivalent reformulations.

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